Ans. (b) : Le the third number Z = 100

$\therefore$ x = 100 - 20 = 80

y = 100 - 28 = 72

Required %

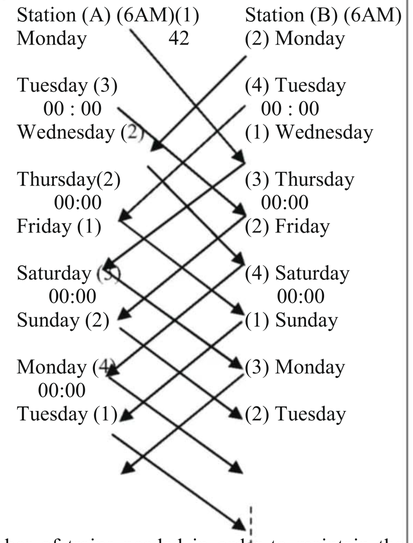
$= \frac{(80-72)}{80} \times 100$

= 10%

38. A daily train is to be introduced between station A and station B starting from each end at 6 am and the journey is to be completed in 42 hours.
What is the number of trains needed in order to maintain the Shuttle Service?

(a) 2 (b) 3
(c) 4 (d) 7

Ans. (c) :



The number of trains needed in order to maintain the shuttle service is 4.

39. A piece of tin is in the form of a rectangle having length 12 cm and width 8 cm. This is used to construct a closed cube. The side of the cube is ?

(a) 2 cm (b) 3 cm
(c) 4 cm (d) 6 cm

Ans. (c) :

Let the side of cube = a

then,

$6a^2 = 12 \times 8$

$a^2 = 2 \times 8 = 16$

$a^2 = 16 \Rightarrow a = 4$ Cm.

40. In a question paper there are five questions to be attempted and answer to each questions has two choices-True (T) or False (F). It is given that no two candidates have given the answers to the five questions in an identical sequence. For this to happen the maximum number of candidates is:

(a) 10 (b) 18
(c) 26 (d) 32

Ans. (d) :

the maximum number candidates =

$\Rightarrow [5C_1 + 5C_2 + 5C_3 + 5C_4 + 5C_5 + 1]$

$\Rightarrow \left[\frac{\lfloor 5}{\lfloor 1 \lfloor 5-1} + \frac{\lfloor 5}{\lfloor 2 \lfloor 5-2}\right] + \frac{\lfloor 5}{\lfloor 3 \lfloor 5-3} + \frac{\lfloor 5}{\lfloor 4 \lfloor 5-4} + \frac{\lfloor 5}{\lfloor 5 \lfloor 5-5} + 1\right]$

$\Rightarrow \left[\frac{\lfloor 5}{\lfloor 1 \lfloor 4} + \frac{\lfloor 5}{\lfloor 2 \lfloor 3} + \frac{\lfloor 5}{\lfloor 3 \lfloor 2} + \frac{\lfloor 5}{\lfloor 4 \lfloor 1} + \frac{\lfloor 5}{\lfloor 5 \lfloor 0} + 1\right]$

$\Rightarrow \frac{5 \times \lfloor 4}{\lfloor 4} + \frac{5 \times 4 \times \lfloor 3}{1 \times 2 \times \lfloor 3} + \frac{5 \times 4 \times \lfloor 3}{1 \times 2 \lfloor 3} + \frac{5 \times \lfloor 4}{\lfloor 4} + 1 + 1$

$\Rightarrow$ 5+5×2 +5×2 +5×1+1+1

= 5+10+10+5+1+1

= 32

Directions for the following 8 (eight) items:

Read the following eight passages and answer the items that follow each passage. Your answers to these items should be based on the passages only.

Passage -1

By killing transparency and competition, crony capitalism is harmful to free enterprise, opportunity and economic growth. Crony capitalism, where rich and the influential are alleged to have received land and natural resources and various licenses in return for payoffs to venal politicians, is now a major issue to be tackled. One of the greatest dangers to growth of developing economies like India is the middle-income where crony capitalism creates oligarchies that slow down the growth

41. Which among the following is the most logical corollary to the above passage ?

(a) Launching more welfare schemes and allocating more finances for the current schemes are urgently needed
(b) Efforts should be made to push up economic growth by other means and provide licences to the poor
(c) Greater transparency in the functioning of the government and promoting the financial inclusion are needed at present
(d) We should concentrate more on developing manufacturing sector than service sector

Ans. (c) : The most logical corollary to the above passage is that greater transparency in the functioning of the government and promoting the financial inclusion are needed at present. Transparancy will reduce corruption and those who take undue advantage in collusion with the government will be curbed.

Passage 2

Climate adaptation may be rendered ineffective if policies are not designed in the context of other development concerns. For instance, a comprehensive strategy that seeks to improve food security in the context of climate change may include a set of coordinated measures related to agricultural extension, crop diversification, integrated water and pest management and agricultural information services. Some of these measures may have to do with climate changes and others with economic development.

42. What is the most logical and rational inference that can be made from the above passage?

(a) It is difficult to pursue climate adaptation in the developing countries
(b) Improving food security is a far more complex. issue than climate adaptation
(c) Every developmental activity is directly or indirectly linked to climate adaptation
(d) Climate adaptation should be examined in tandem with other economic development options.

Ans. (d) : The most logical and rational inference than can be made from the above passage is that climate adaptation should be examined in relation with other economic development options.

Passage -3

Understanding of the role of biodiversity in the hydrological cycle enables better policy-making. The term biodiversity refers to the variety of plants, animals, microorganisms, and the ecosystems in which they occur. Water and biodiversity are interdependent. In reality, the hydrological cycle decides how biodiversity functions. In turn, vegetation and soil drive the movement of water. Every glass of water we drink has, at least in part, passage through fish, trees, bacteria, soil and other organisms. Passing through these ecosystems, it is cleansed and made fit for consumption .The supply of water is a critical service that the environment provides.

43. Which among the following is the most critical inference that can be made from the above passage?

(a) Biodiversity sustains the ability of nature of recycle water
(b) We cannot get potable water without the existence of living organisms
(c) Plants, animals and microorganisms continuously interact among themselves
(d) Living organisms could not have come into existence without hydrological cycle.

Ans. (a) : The most critical inference that can be made from the above passage is that biodiversity sustains the ability of nature of recycling water and water is purified through biodiversity. There is a difference between artificial purified water and naturally purified water.

Passage -4

In the last decade, the banking sector has been restructured with a high degree of automation and products that mainly serve middle-class and upper middle-class society. Today there is need for a new agenda for the banking and non-banking financial services that does not exclude the common man.

44. Which one of the following is the massage that is essentially implied in the above passage?

(a) Need for more automation and more products of bank
(b) Need for a radical restructuring of our entire public finance system
(c) Need to integrate banking and non-banking institutions
(d) Need to promote financial inclusion

Ans. (d) : The message that is essentially implied in the above passage is that there is a need to promote financial inclusion. Financial inclusion will increase the liquidity of money and if money reaches to the people easily then people will automatically reach the bank.

Passage -5

Safe and sustainable sanitation in slums has immeasurable benefits to women and girls in terms of their health, safety, privacy and dignity. However, women do not feature in most of the schemes and policies on urban sanitation. The fact that even now the manual scavenging exists only goes to show that not enough had been done to promote pour-flush toilets and discontinue the use of dry latrines. A more sustained and rigorous campaign needs to be launched towards the right to sanitation on very large scale. This should primarily focus on the abolition of manual scavenging.

45. With reference to the above passage, consider the following statements :

1. Urban sanitation problems can be fully solved by the abolition of manual scavenging only
2. There is a need to promote greater awareness on safe sanitation practices in urban areas

Which of the statements given above is /are correct ?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (b) : There is a need to encourage more awareness of safe clearing practices is urban areas. The above statement is true in that context of the passage.

Passage – 6

To Understand the nature and quantity of Government proper for man, it is necessary to attend to his character. As nature created him for social life, she fitted him for the station she intended, In all cases she made his natural wants greater than his individual owers. No one man is capable, without the aid to society , of spplying his own wants; and those wants, acting upon every individual, impel the whole of them into society.

46. Which among the following is the most logical and rational inference that can be made from the above passage?

(a) Nature had created a great diversity in human society
(b) Any given human society is always short of its wants
(c) Social life is a specific characteristic of man
(d) Diverse natural wants forced man towards social system.

Ans. (d) : The most logical and rational inference that can be drawn from the above passage is that various natural needs have compelled human beings towards the social system.